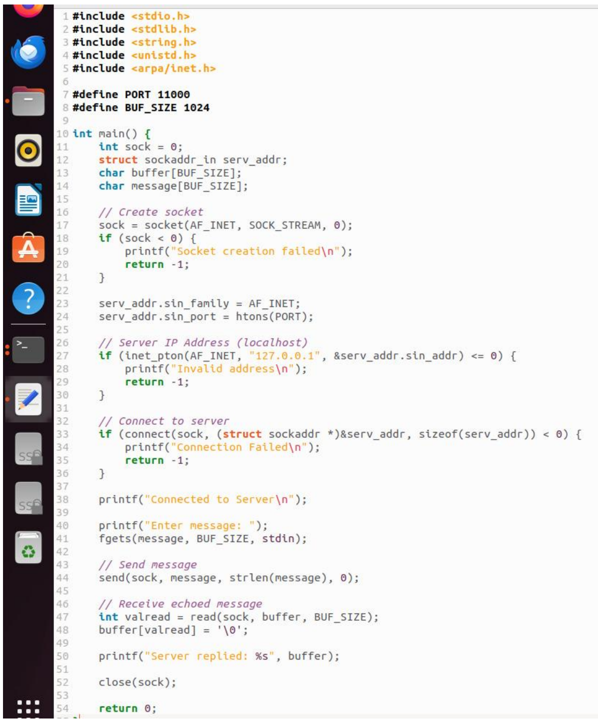
NAME : LAVANYA TIWADE

PRN : 24070521130

SECTION : B2



```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 11000
#define BUF_SIZE 1024

int main() {
    int sock = 0;
    struct sockaddr_in serv_addr;
    char buffer[BUF_SIZE];
    char message[BUF_SIZE];

    // Create socket
    sock = socket(AF_INET, SOCK_STREAM, 0);
    if (sock < 0) {
        printf("Socket creation failed\n");
        return -1;
    }

    serv_addr.sin_family = AF_INET;
    serv_addr.sin_port = htons(PORT);

    // Server IP Address (localhost)
    if (inet_pton(AF_INET, "127.0.0.1", &serv_addr.sin_addr) <= 0) {
        printf("Invalid address\n");
        return -1;
    }

    // Connect to server
    if (connect(sock, (struct sockaddr *)&serv_addr, sizeof(serv_addr)) < 0) {
        printf("Connection Failed\n");
        return -1;
    }

    printf("Connected to Server\n");

    printf("Enter message: ");
    fgets(message, BUF_SIZE, stdin);

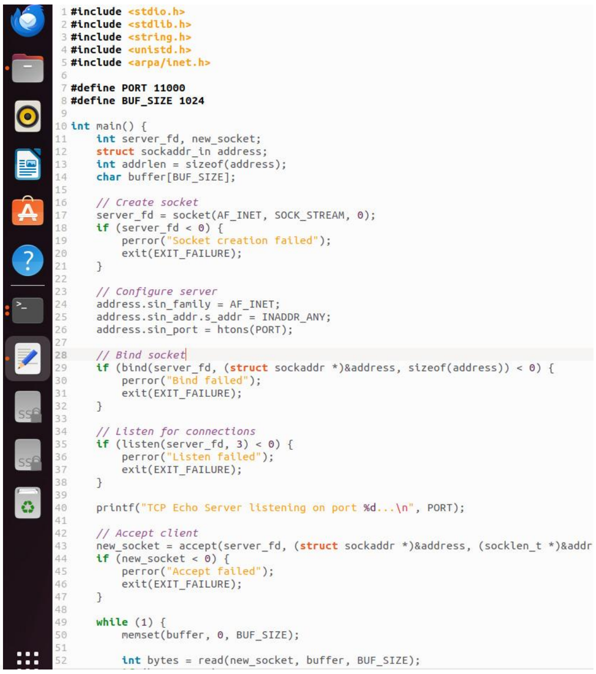
    // Send message
    send(sock, message, strlen(message), 0);

    // Receive echoed message
    int valread = read(sock, buffer, BUF_SIZE);
    buffer[valread] = '\0';

    printf("Server replied: %s", buffer);

    close(sock);

    return 0;
```



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#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 11000
#define BUF_SIZE 1024

int main() {
    int server_fd, new_socket;
    struct sockaddr_in address;
    int addrlen = sizeof(address);
    char buffer[BUF_SIZE];

    // Create socket
    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    if (server_fd < 0) {
        perror("Socket creation failed");
        exit(EXIT_FAILURE);
    }

    // Configure server
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);

    // Bind socket
    if (bind(server_fd, (struct sockaddr *)&address, sizeof(address)) < 0) {
        perror("Bind failed");
        exit(EXIT_FAILURE);
    }

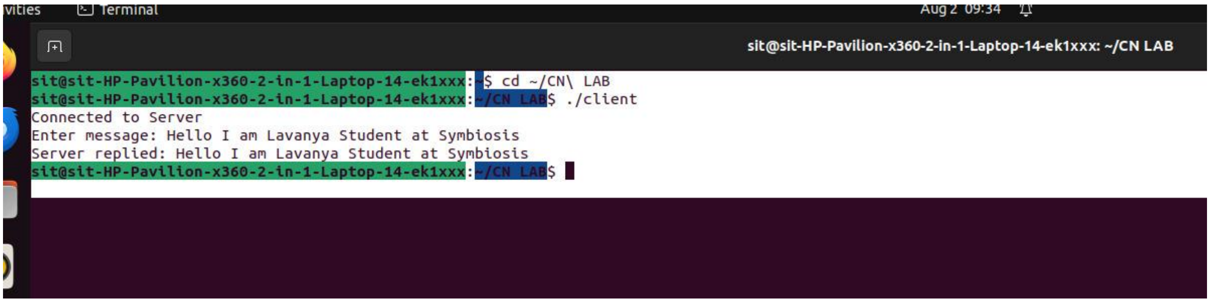
    // Listen for connections
    if (listen(server_fd, 3) < 0) {
        perror("Listen failed");
        exit(EXIT_FAILURE);
    }

    printf("TCP Echo Server listening on port %d...\n", PORT);

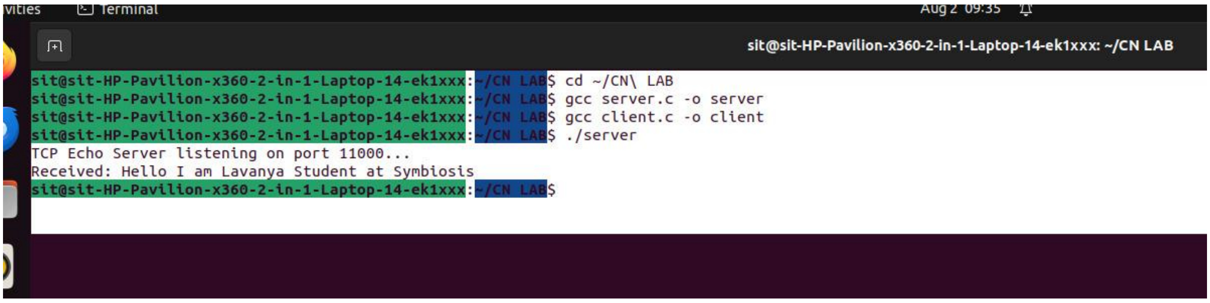
    // Accept client
    new_socket = accept(server_fd, (struct sockaddr *)&address, (socklen_t *)&addr
    if (new_socket < 0) {
        perror("Accept failed");
        exit(EXIT_FAILURE);
    }

    while (1) {
        memset(buffer, 0, BUF_SIZE);

        int bytes = read(new_socket, buffer, BUF_SIZE);
```



```
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~$ cd ~/CN\ LAB
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$ ./client
Connected to Server
Enter message: Hello I am Lavanya Student at Symbiosis
Server replied: Hello I am Lavanya Student at Symbiosis
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$ 
```



```
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$ cd ~/CN\ LAB
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$ gcc server.c -o server
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$ gcc client.c -o client
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$ ./server
TCP Echo Server listening on port 11000...
Received: Hello I am Lavanya Student at Symbiosis
sit@sit-HP-Pavilion-x360-2-in-1-Laptop-14-ek1xxx:~/CN LAB$
```